

Litesh Perumalla

AI Engineer

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SUMMARY

Data Science graduate with strong foundations in Python, R, SQL, and applied machine learning, targeting AI Engineer roles. Built end-to-end projects including a RAG-based LLM application, multi agent architectures, predictive models, and data pipelines using real-world datasets. Demonstrated ability to communicate insights and support users and collaborating on cross-functional projects. Looking to help teams design, evaluate, and scale AI solutions that improve decision-making.

EDUCATION

University of North Texas | *Master of Science, Data Science (GPA: 3.9)*

Jan 2024 - Dec 2025

EMPLOYMENT EXPERIENCE

crypt0nest.io | *AI Engineer Intern (Team Lead)*

May 2025 - Aug 2025

- Led development of a predictive crypto portfolio system, integrating time-series forecasting models and API-driven pipelines to improve accuracy and scalability.
- Engineered volatility-based features and performed back-testing with mean-variance optimization, enhancing model robustness and reliability.
- Designed and maintained scalable NoSQL databases for real-time and historical crypto data, improving data accessibility and system performance.
- Architected a multi-agent LLM-based system with RAG and prompt engineering, enabling automated reasoning, decision-making, and hybrid AI integration.
- Built and deployed an MCP server-client framework interfacing with LLMs, APIs, and databases, while leading a team of 5+ interns to automate workflows and increase decision efficiency by 30%.

PROJECT EXPERIENCE

Infra Mind – AI Powered Infrastructure Platform

May 2025 – Dec 2025

- Delivered an AI Infrastructure advisory platform with 10+ agents that analyze assessments, compare cloud providers and generate strategic recommendations.
- Built 5+ endpoints like executive and compliance features with role based and assessment level authorization.
- Architected a multi-agent system with 10+ agents to automate infrastructure provisioning and compliance monitoring, reducing manual overhead and improving scalability.
- Collaborated across teams to integrate enterprise-grade security and rollback automation, ensuring compliance and system resilience.
- Utilized tools such as React.js, Fast API, MongoDB, Docker, and Azure OpenAI to deliver a robust, scalable.

Smart Tutor AI – Personalized Teaching Support

Jan 2025 – May 2025

- Engineered to significantly improve student engagement and learning outcomes by providing instant, 24/7 academic support. The application delivers context-aware answers from course materials, offers on-demand quiz generation for self-assessment, and clarifies complex topics, fostering deeper understanding and greater learning efficiency.
- A production level personalized AI tutoring system using AWS services for storage, inference, embeddings, monitoring, security, and improving response.
- Developed an end-to-end metadata-driven CRAG and Self-RAG pipeline, ingesting course materials and multi format parsing with hierarchical semantic chunking, faster similarity searches, accuracy, citations, evaluations, and hybrid storage for scalability.
- Designed and implemented a User Friendly full-stack application featuring modular services, API's, user authentication, chat history, session management, quizzes, feedback collection, and a research mode for external sources.
- Architected and implemented a comprehensive evaluation and observability framework to continuously measure and improve the RAG pipeline's quality and performance. Integrated Langfuse for end-to-end LLM tracing and Prometheus for system metrics, enabling the tracking of key retrieval/generation metrics (nDCG, Faithfulness, Precision@k) and operational KPIs like query latency and cost.

TECHNICAL SKILLS

- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, MLflow.
- **MLOps & Infrastructure:** Kubernetes (EKS), Docker, Jenkins, GitLab CI, Terraform, CI/CD, CORS.
- **Cloud Platforms:** AWS (EC2, S3, Lambda, SageMaker, EKS, Bedrock, DynamoDB, S3 vectors, IAM, Secret Manager), GCP, Azure
- **Programming:** Python, SQL, R
- **Databases:** PostgreSQL, MongoDB, Redis, NoSQL, Dynamo DB, vector search.
- **Monitoring & Logging:** Prometheus, Grafana, AWS CloudWatch, Lang fuse,
- **Version Control:** Git, GitHub, Gitlab Actions.